Example 7.50

Find the intervals of monotonicity and hence find the local extrema for the function $f(x) = x^2 - 4x + 4$.

E3I2Y4

Solution

We have,

$$f(x) = (x-2)^2, \text{ then}$$
$$f'(x) = 2(x-2) = 0 \text{ gives } x = 2 .$$

The intervals of monotonicity are $(-\infty, 2)$ and $(2, \infty)$. Since $f'(x) < 0$, for $x \in (-\infty, 2)$ the function $f(x)$ is strictly decreasing on $(-\infty, 2)$. As $f'(x) > 0$, for $x \in (2, \infty)$ the function $f(x)$ is strictly increasing on $(2, \infty)$. Because $f'(x)$ changes its sign from negative to positive when passing through $x = 2$ for the function $f(x)$, it has a local minimum at $x = 2$. The local minimum value is $f(2) = 0$. ■

Example 7.51

Find the intervals of monotonicity and hence find the local extrema for the function $f(x) = x^{\frac{2}{3}}$.

Solution

We have, $f(x) = x^{\frac{2}{3}}$, then $f'(x) = \frac{2}{3}x^{-\frac{1}{3}} = \frac{2}{3x^{\frac{1}{3}}}$. $f'(x) \neq 0 \quad \forall x \in \mathbb{R}$ and $f'(x)$ does not exist at $x = 0$. Therefore, there are no stationary points but there is a critical point at $x = 0$.

Interval	(-∞, 0)	(0, ∞)
Sign of $f'(x)$	−	+
Monotonicity	strictly decreasing	strictly increasing

Table 7.5

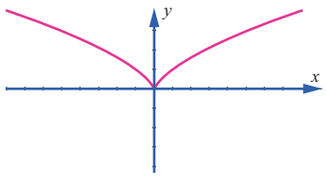


Fig.7.21

Because $f'(x)$ changes its sign from negative to positive when passing through $x = 0$ for the function $f(x)$, it has a local minimum at $x = 0$ .The local minimum value is $f(0) = 0$. Note that here the local minimum occurs at a critical point which is not a stationary point. ■

Example 7.52

Prove that the function $f(x) = x - \sin x$ is increasing on the real line. Also discuss for the existence of local extrema.

Solution

Since $f'(x) = 1 - \cos x \geq 0$ and zero at the points $x = 2n\pi$, $n \in \mathbb{Z}$ and hence the function is increasing on the real line.

Since there is no sign change in $f'(x)$ when passing through $x = 2n\pi$, $n \in \mathbb{Z}$ by the first derivative test there is no local extrema. ■

Example 7.53

Discuss the monotonicity and local extrema of the function

$$f(x) = \log(1+x) - \frac{x}{1+x}, x > -1 \text{ and hence find the domain where, } \log(1+x) > \frac{x}{1+x}.$$

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